

Replication Paper: Heterogeneous Spillovers of Housing Credit Policy Pidkuyko 2023

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1 Abstract

We replicate Pidkuyko 2023, which studies the spillovers from government intervention in the mortgage market on households' consumption. We confirm his results regarding the heterogeneity of effects through replication. Then we extend his instrument variable-local projection models by investigating heterogeneous effects in three new groups: race, education level, and family size. We find that this policy has a stronger impact for White households and households with college degrees. The results suggest disparities in the distribution of impacts from government intervention in the mortgage market.

2 Introduction

In the United States, the largest asset many people will obtain is a house. Personal incentives, government policy, and the financial system have all developed to facilitate the reality that home ownership is a major source of stability and security. The majority of household borrowing is residential mortgage debt. Through Government-Sponsored Enterprises (GSEs) and Government Agencies, the US government actively engages in the secondary mortgage market. At one point, GSEs held nearly 20% of the overall mortgage debt in the economy. Other research has documented that secondary market purchases by GSEs boost mortgage lending, lower mortgage rates, and influences prices in other financial markets (Fieldhouse et al., 2018). Pidkuyko analyzes how these effects impact household consumption, which makes up the largest component of GDP.

Pidkuyko's analysis shows that there is a significant positive response to GSE secondary mortgage market activity in consumer spending. He then explores the heterogeneity of these effects based on different housing tenure statuses (i.e., mortgagors, renters, and outright homeowners) and age groups (i.e. young, middle aged, and old). This paper finds that there are, in fact, heterogeneous effects from this secondary market activity, with the greatest effects for mortgagors and the young.

For his analysis, Pidkuyko focuses on expansionary credit policy shocks, or an increase in

agency purchases. Specifically, he defines the shock as a one percent increase in net purchases by Fannie Mae (FNMA) and Freddie Mac (FHLMC) anticipated two quarters ahead of time. By their nature as a government policy tool, these purchase increases are endogenously related to the business cycle. As a result, Pidkuyko uses narrative news shocks to identify policies orthogonal to the business cycle which act as instruments. For instance, a shock orthogonal to the business cycle could be the announcement of a long-term objective to increase home ownership. Pidkuyko then uses these instruments in an IV-local projection to identify the shocks and create impulse response functions. This identification strategy works because it facilitates exogenous shock identification and allows for a predictive model that is not overly susceptible to misspecification across time like VARs can be.

In addition to the empirical analysis of the local projection, Pidkuyko conducts a theoretical exercise relying on a Huggett (1996) heterogeneous agent life-cycle model. The purpose of this exercise is to parse the mechanisms of the empirical results. While the empirical results are compelling, they do not explain what component of the expansionary credit policy increases household consumption. Through his theoretical model, he concludes that the main mechanisms that increases household consumption are the decreases on mortgage rates, the spread between mortgage and interest rates, refinancing incentives, and the increase of disposable cash flow.

In this paper, we replicate Pidkuyko's empirical analysis and extend it by exploring heterogeneous treatment effects in other groups. Specifically, we compare differences in effects based on race, constructing models assessing effects on Black vs. white vs. other races. Then we compare differences in education levels, constructing models assessing effects on households where the head does not have a high school degree vs. households where the head has a high school degree vs. households where the head has a college degree. Finally, we compare the differences between households of different family sizes: single households, two-person households, and families with three or more members. We find that the expansionary credit policy leads to a significant increase in consumption for college graduates, white households, and all family types, though the biggest impact is on two person households and families. There is a smaller increase in consumption for Black households, but it is not significant. These findings suggest expansionary credit policy is a tool that does not aid the population uniformly and varies among important demographic dividers.

We forego the replication of their theoretical modeling because it is outside the scope of this course and non-essential to the broad findings of this paper.

3 Data

The data in this paper consist of a number of macro variables used as part of the local projections, consumer expenditure aggregations, consumer finance data used for the Huggett (1996) model, and a series of news shocks used as IVs for the local projection. Appendix A

of the original paper contains additional details on specific data sources and calculations.

The macro variables are fairly standard variables used in macroeconomic work. They are found primarily on FRED, though some were created using the researcher’s calculations. The consumer expenditure aggregations are computed using interview data from the Consumer Expenditure Survey, which can be accessed on the Bureau of Labor Statistics website and ICPSR. The researcher uses several do files to aggregate raw interview data into a compiled data set. Unfortunately, component do files needed for the aggregation are omitted from the researcher’s replication package, so we were unable to replicate the construction using raw data. Fortunately, their compiled data is included in the replication package, which we use to construct new groups for our extension. The consumer finance data is from the Survey of Consumer Finances available through the Federal Reserve. Finally, the series of news shocks comes from a dataset constructed by Fieldhouse and Mertens (2017). The final datasets used for this paper and our replication ranges from 1980 to 2007 and are quarterly time series.

For our extension, we use the compiled consumption data to create several binary variables related to race (Black, white, and other), educational attainment (not a high school grad, high school grad, and college grad), and family size (single, two-person, and family). Using the indicator variables, we then collapse our observations into mean nondurable consumption expenditures in each quarter from 1980-2007 based on their groups. The resulting wide time series serves as the outcome of interest in our local projections.

4 Empirical Framework

The original paper uses instrumental variable local projection estimation to create models that estimate the effect of a one percent net increase in net purchases by FNMA and FHLMC on nondurable consumption. Models are created that look at the population at large and in two sub groups to allow for the identification of heterogeneous effects. The two groups are differentiated by housing tenure status and age. Housing tenure status consists of renters, mortgagors (homeowners with a mortgage), and owners (homeowners who do not have a mortgage). Age consists of young (25 to 34 years old), middle aged (between 35 and 64 years old), and old (between 65 and 75 years old). Because expansionary credit policy is endogenous to downturns in the business cycle, news shocks are used as IVs so the shock is orthogonal to business cycle expectations.

We extend this estimation by investigating three more subgroups for heterogeneous effects. The three groups are differentiated by race, education level, and family size. The race subgroup consists of Black, white, and other, based on the race of the head of household. The education subgroup consists of no high school degree, a high school degree, and a college degree, based on the highest educational attainment of the head of household. Finally, the family size subgroup consists of single households where only one person lives, two-person households where two people live, and families where three or more people live.

The main estimating equation is

$$\log(y_{t+k}) - \log(y_{t-1}) = a_k + b_k\left(\frac{4}{2} \times \frac{\sum_{j=0}^2 p_{t+j}}{x_t}\right) + d_k(L)Z_{t-1} + u_{t+k}$$

where the term in parentheses denotes net agency purchases made over a two quarter period divided by annualized originations X_t . The Z vector is a set of control variables used by Fieldhouse et al. (2018): the lagged growth rates of the core PCE price index, a nominal house price index, and total mortgage debt, the log level of real mortgage originations, housing starts, and lags of the 3-month T-bill rate, the 1- year Treasury rate, the conventional mortgage interest rate, and the BAA-AAA corporate bond spread. a_k serves as a constant and u_{t+k} is an unobserved error term.

The key assumption of this empirical framework is that the news shocks used as an instrument are orthogonal to the business cycle, allowing us to remove the endogeneity of the shock. This is plausible given the longer term policies that represent the orthogonal shocks are unrelated to near term business cycle fluctuations. Additional assumptions include the IV-exclusion, i.e. the IV does not affect the outcome variable directly, stationarity which is likely after differencing our outcomes, proper lag structure which is likely satisfied by the four quarters (one year) used by the researcher, and that we can estimate heterogeneous treatment effects the same way for each group.

5 Theoretical Framework

In this section we will discuss two theoretical frameworks. First, we will discuss the theory that underscores the empirical hypothesis and findings of Pidkuyko. Then, we will briefly discuss the Huggett (1996) model that Pidkuyko uses to parse the components of expansionary credit policy that lead to the increase in household consumption.

The key theoretical framework of this paper is premised on the idea that government purchases in the secondary mortgage market can help to stimulate the economy by increasing GDP by way of household consumption. When business cycles enter a trough, ESGs can increase their activity in the secondary mortgage market, stimulating the mortgage market at large. When primary mortgage lenders see that the government is purchasing mortgages, they feel more comfortable providing mortgages to borrowers. The increased demand created in the secondary market helps offset lending risk in the primary market. Like playing hot potato, there is less risk when there are more hands to be left holding the hot potato. Additionally, the government often has more risk tolerance when it comes to taking on loans, further decreasing risk in the primary market.

When primary mortgage market lenders have decreased risk, they can supply more mortgages at a lower price. In this case the price is the mortgage rate. As a result, mortgage borrowers are able to purchase more mortgages at a lower price or rate. Often, government pulls this policy lever in a business cycle downturn. As a result, the actual rates and

quantity of mortgages may still worsen, but they will be better relative to other economic conditions. This may incentivize mortgage borrowing in a manner that ultimately frees up cash for household consumption, a major component of GDP.

When mortgage rates are cheaper, more households may be able to purchase a home or may choose to refinance their home. Refinancing as well as the stability of a mortgage payment can help increase consumer spending. Households no longer need to anticipate unknown raises to their rent. After refinancing, households have access to new disposable income. As the empirical findings show, however, the effects are likely to be heterogeneous. Individuals who outright own their homes are less affected to changes in mortgage rates. Unless the terms are so good that they make a second mortgage a good investment, they are unlikely to change their habits based on differences in mortgage rates. Renters, too, are more insulated from changes in mortgage rates. However, their rent may become more stable if their landlord refinances for a lower rate. Additionally, lower rates may inspire them to purchase a house or to save more for a house. Despite this heterogeneity, the fundamental idea is that decreasing the prices of mortgages frees up cash for additional spending, ultimately spurring on the economy when there are troughs.

The Huggett (1996) model is a heterogeneous agent life-cycle model with uninsurable risk, endogenous housing choice, and aggregate mortgage market shocks. This model allows the researcher to explain the underlying mechanics of empirical findings. This dynamic model is outside the scope of this course. However, it is worth discussing the findings of this model. This model demonstrates that the increase in consumption is the result of lower mortgage rates that improve refinancing options, ultimately decreasing mortgage payments and increasing consumption among mortgagors. Expansionary credit policy shocks push down mortgage rates. Lower rates present refinancing options for mortgagors, reducing spending. Some of this relief is used for consumption.

This mechanism can also raise home prices as demand increases and home-ownership, which decreases supply. However, the effects of the price increase are overpowered by the decreased mortgage rates, the spread between interest and mortgage rates, refinancing incentives, and disposable cash flows.

6 Results

This paper finds that expansionary credit policy in the form of a one-percent net increase in net purchases by GSEs results in an increase in nondurable consumption across the population. This effect occurs as early as two quarters after the shock, but it is not statistically significant until quarter four. The increase persists for two years and then returns to baseline in quarter nine. The increase in nondurable consumption spending is approximately 0.05%, so it is relatively small. However, this suggests that this policy lever does have a meaningful impact when it comes to offsetting business declines.

Additionally, this paper finds there are heterogeneous effects as a result of this expansionary credit policy. The effects differ between different housing tenure types and age groups. Among the three housing tenure types, mortgagors face the greatest impact, which is very similar to the overall effect. They have 0.05% increase in nondurable consumption expenditure that becomes statistically significant in quarter three and persists through quarter eight and returns to baseline in quarter nine. The owner group appears to have no consistent nor significant impact from the policy. The renter group appears to have a positive impact on nondurable consumption expenditure, however, it is smaller than mortgagors and never statistically significant. In the age group, the young and middle-aged groups have positive increases in their consumption, with young having the greatest impact. Both revert to baseline in quarter nine and are statistically significant by quarter four. The old group appears to have no consistent effect from the policy impact.

These findings make sense. First, mortgagors are most likely to be impacted by these expansionary credit policy shocks. With the introduction of cheaper mortgages, they are able to refinance and divert some of their newly available cash to consumption. The positive, yet smaller effect on renters likely has to do with a spillovers from decreased housing costs. Their landlords are less likely to raise rents as drastically in the short term, so they can safely increase their spending. Outright owners of their homes are unaffected because their housing costs are not impacted by mortgages. An exception may be if mortgage rates become so low that it could be beneficial to take out a second mortgage to provide cash for other savings, investment, or spending.

Similarly, we expect the young and the middle-aged to be affected given where they stand in the housing market. The young and middle-aged are more likely to pursue mortgage refinancing because their mortgages are almost certainly far from maturity. The old will either have mortgages nearing their maturity or may have paid them off already, meaning there is little benefit from refinancing opportunities.

Our replication of the researcher’s models was successful. We were able to create IRFs that mirrored the researchers using their data and replication package. Unfortunately, they were missing components of their code that would have allowed us to reconstruct their dataset.

Our extension provided interesting results regarding heterogeneous effects as well. Using the researcher’s IV-LP models with new consumption time-series data, we created IRFs for each group type in our three groups. See all IRFs below.

Within the race group, only white households (based on head of household) received a statistically significant effect from the expansionary credit policy shock. A positive impact began by quarter two, became significant in quarter three, and persisted for eight quarters. At its peak, the increase in consumption was nearly 0.1%. Black households saw a smaller, delayed increase in consumption, however it is never statistically significant. Other race households appeared to see no impact from the policy. Given that historical systems were designed to remove housing and credit access to non-white communities, it is not surprising that white

households benefit the most from the expansion of housing credit. This provides evidence that barriers to equal access likely still exist on a racial level and should inform the actions of policy makers.

In the education group, only college educated households received a statistically significant effect from the expansionary credit policy shock. The effect follows a similar path as other subgroups with significant positive effects. However, it is not significant across all periods. High graduate households see a similar positive effect to renters, but it does not surpass significance. Finally, non-high school graduates appear to have no discernible impact from the policy. This result is not surprising given general earnings and class differentials based on education levels. More educated households may be more likely to own a house and navigate opportunities for refinancing. Additionally, more educated families may have budgets that are generally less tight, allowing them to more easily expand their consumption when they receive relief. This provides evidence that families that have lacked access to or achievement in education do receive the same relief from this policy lever. Policy makers should consider this because lower educated and lower class individuals are often the hardest hit in recessionary periods. Future research could explore the benefits of policy to expand their access and achievement in education, improve attainment of home-ownership, and provide effects more targeted to this subgroup.

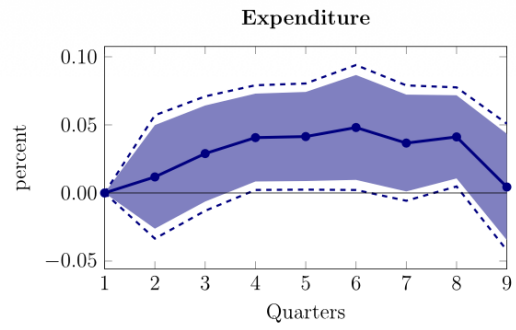
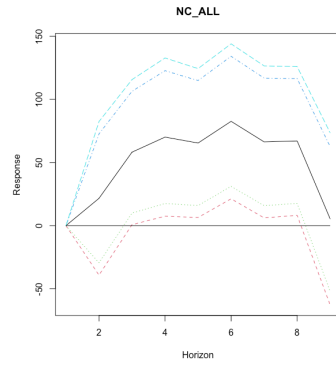
Finally, in the family size group, all subgroups had periods of statistically significant positive impacts on consumption, suggesting the variation of effects based on family size is less pronounced than other groups. That said, two person households have a much greater magnitude of impact compared to the other two groups. Family or three-plus person households also have a stronger impact than single households. We suppose that two-family households have less expenses overall and therefore more room to easily expand their consumption when given relief. Two-person and family sized households we suspect are more likely to be mortgagors than single households who are likely to be renters or outright owners (if old).

Our extension illustrates that expansionary credit policy shocks have heterogeneous effects along several demographic lines that raise equity concerns. It is important for policy makers to understand these different effects to ensure they are aware of the distribution of government provided relief.

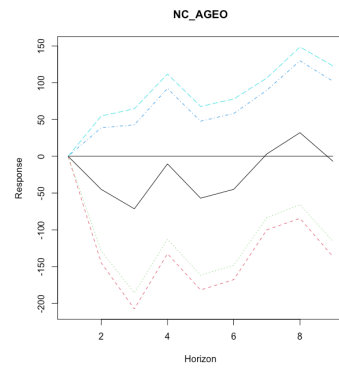
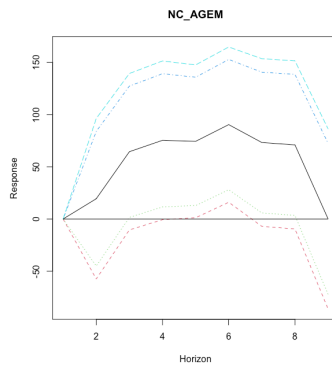
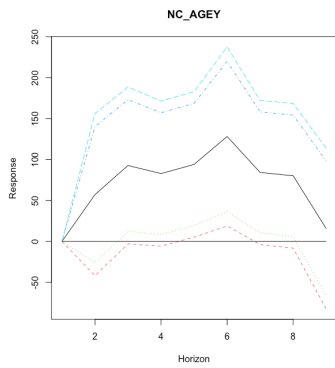
7 Conclusion

This paper uses IV-Local Projections to explore the impact of expansionary credit policy shocks. It finds that this policy can be used to help stimulate GDP growth by way of consumption at an aggregate level. However, the consumption increases are not distributed evenly throughout the economy. The original researcher finds that this policy lever effects the young and middle aged as well as mortgagors more than other subgroups. Our extension suggests there are additional distributional concerns, primarily along racial and educational levels. This policy seems to provide a positive effect for white households and households

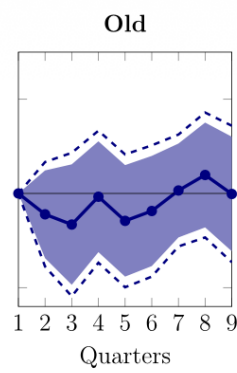
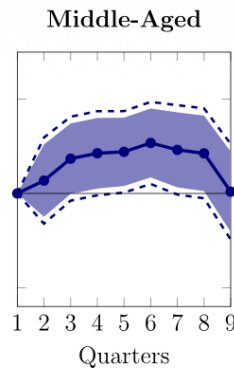
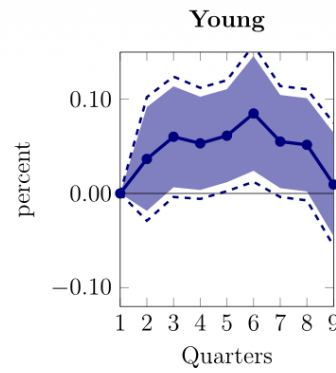
with more education, particularly college degrees. Given that the policy directly affects the mortgage market, it is not surprising that the effect is greatest on people who are most likely to access this market. It remains important for policy makers to understand the effects of their tools.

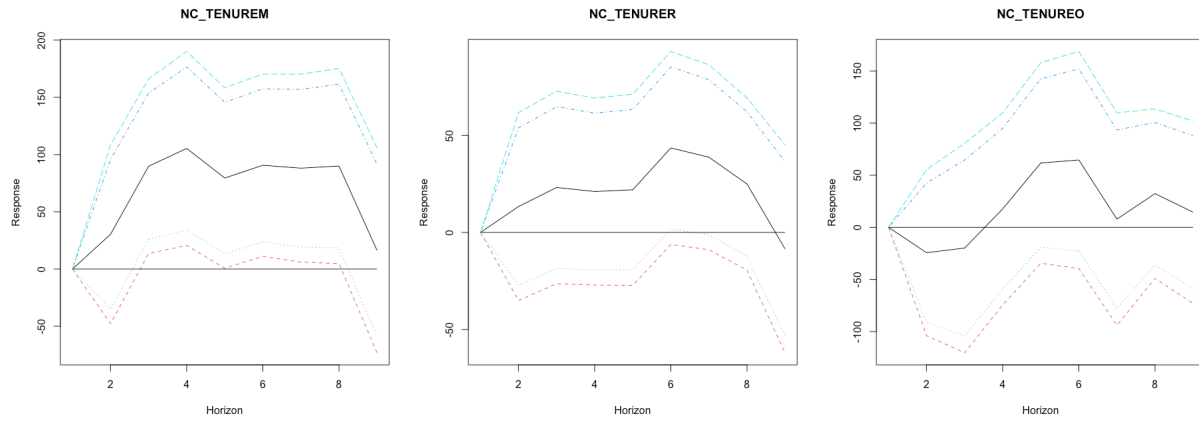


IRF for overall expenditure: replication on right, original on left

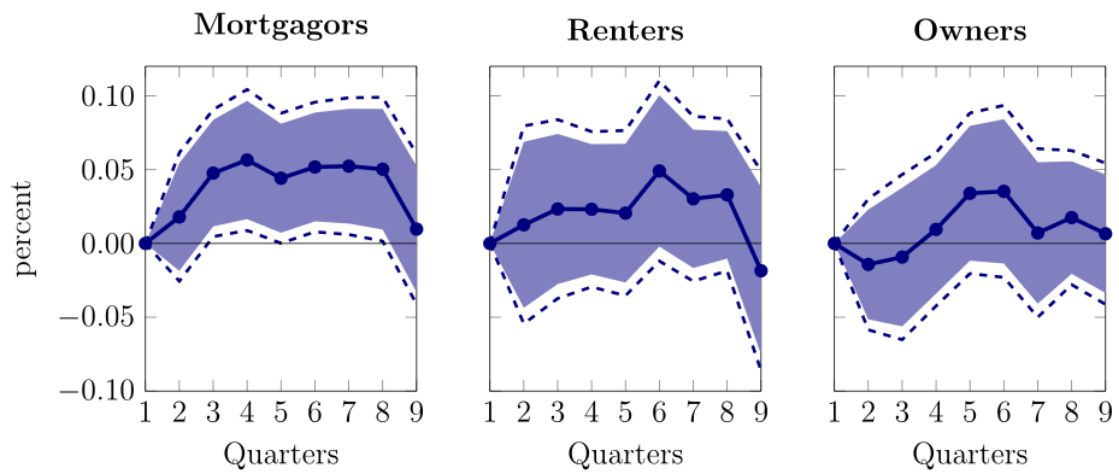


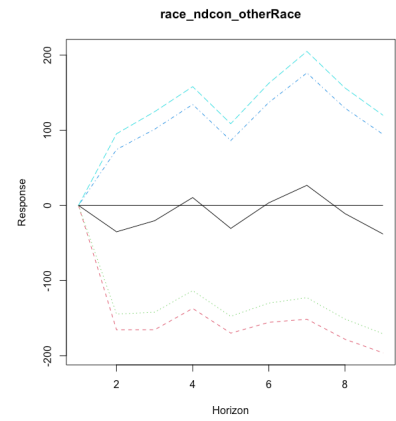
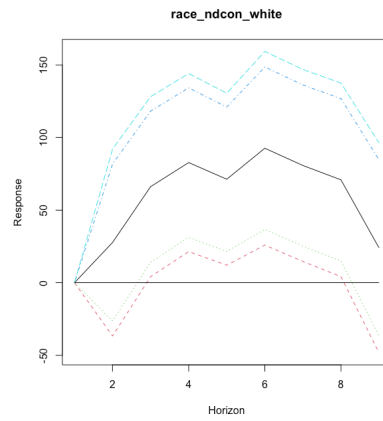
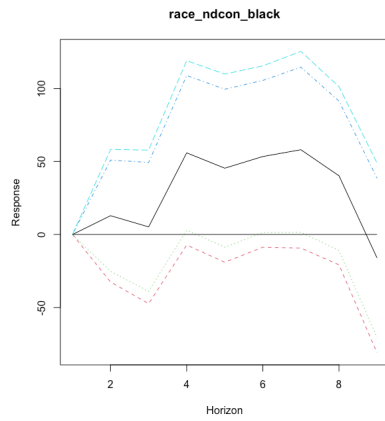
IRFs for age groups: replication above, original below



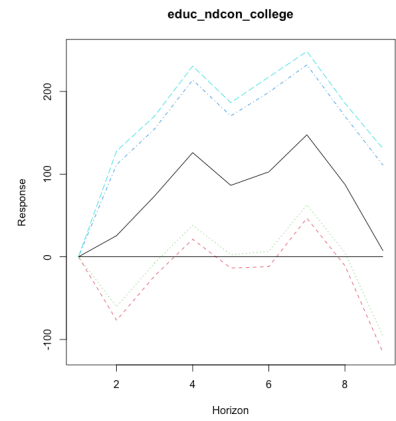
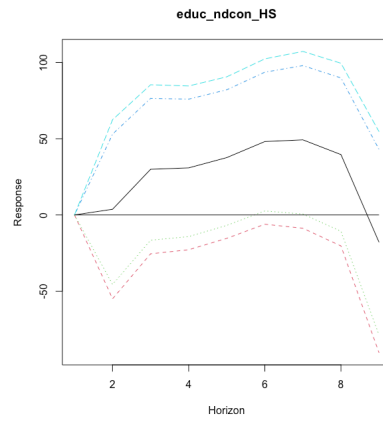
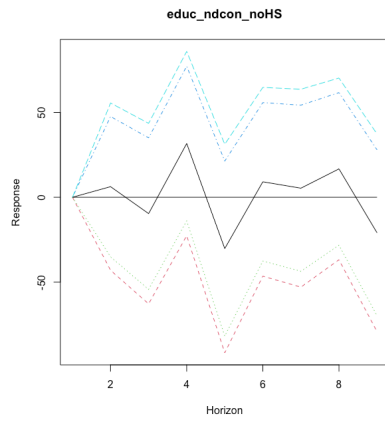


IRFs for housing tenure status: replication above, originals below

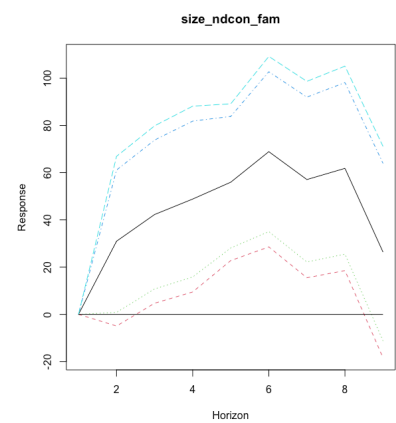
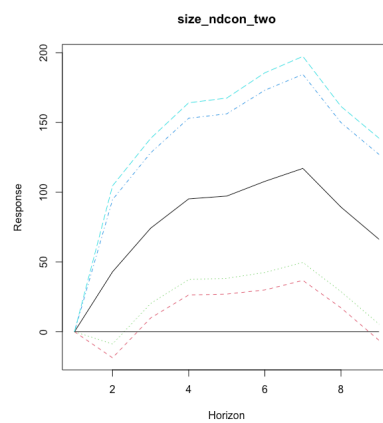
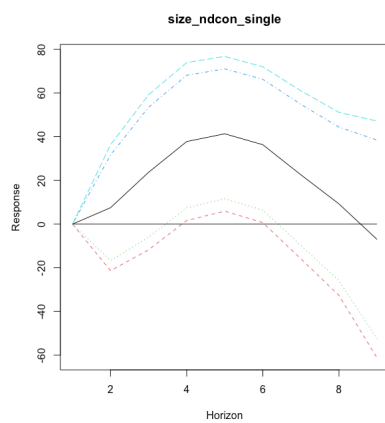




IRFs for race group



IRFs for education level group



IRFs for family size group

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